



Module 15: Database Administration Tools

CPE 011 Database Management System

Slides based on “Database Systems: Design, Implementation, and Management 10th Edition” Chapter 15: Database Administration and Security Slides by Rob, P. and Coronel, C.

The Data Dictionary

- Two main types of data dictionaries:
 - Integrated
 - Standalone
- Active data dictionary is automatically updated by the DBMS with every database access
- Passive data dictionary requires running a batch process
- **Main function: store description of all objects that interact with database**

About the Data Dictionary

- Data dictionary that includes data external to DBMS becomes flexible tool
 - Enables use and allocation of all of an organization's information
- Metadata is often the basis for monitoring database use
 - Also for assigning access rights to users
- DBA uses data dictionary to support data analysis and design

CASE Tools

- Computer-aided systems engineering
 - Automated framework for SDLC
 - Structured methodologies and powerful graphical interfaces
- Front-end CASE tools provide support for planning, analysis, and design phases
- Back-end CASE tools provide support for coding and implementation phases

Typical Case Tools

Typical CASE tool has five components

- Graphics for diagrams
- Screen painters and report generators
- Integrated repository
- Analysis segment
- Program documentation generator

Developing a Data Administration Strategy

- Information engineering (IE) translates strategic goals into data and applications
- Information systems architecture (ISA) is the output of IE process
- Implementing IE is a costly process
 - Provides a framework that includes use of computerized, automated, and integrated tools
- Success of information systems strategy depends on critical success factors
 - Managerial, technological, and corporate culture

The DBA at Work

- Technical tasks handled by the DBA in a specific DBMS:
 - Creating and expanding database storage structures
 - Managing database objects
 - Managing end-user database environment
 - Customizing database initialization parameters
- All DBMS vendors provide programs to perform database administrative tasks

The Default Login

- Must connect to the database to perform administrative tasks
 - Username with administrative privileges
- Oracle automatically creates SYSTEM and SYS user IDs with administrative privileges
- Define preferred credentials by clicking on Preferences link, then Preferred Credentials
- Username and passwords are database-specific

Ensuring that the RDBMS Starts Automatically

- DBA ensures database access is automatically started when computer turned on
 - A service is a Windows system name for a special program that runs automatically
- Part of the operating system
- Database instance: separate location in memory reserved to run the database
 - May have several databases running in memory at the same time

Creating Tablespaces and Datafiles

- Database composed of one or more tablespaces
- Tablespace is a logical storage space
 - Physically stored in one or more datafiles
- Datafile physically stores the database's data
 - Each datafile can reside in a different directory on the hard disk
- Database has 1:M relationship with tablespaces
- Tablespace has 1:M relationship with datafiles

Managing the Database Objects: Tables, Views, Triggers and Procedures

- Database object: any object created by end users
- Schema: logical section of the database that belongs to a given user
 - Schema identified by a username
 - Within the schema, users create their own tables and other objects
- Normally, users are authorized to access only the objects that belong to their own schemas

Managing Users and Establishing Security

- User: uniquely identifiable object
 - Allows a given person to log on to the database
- Role: a named collection of database access privileges
 - Authorizes a user to connect to the database and use system resources
- Profile: named collection of settings
 - Controls how much of a resource a given user can use

Customizing the Database Initial Parameters

- Fine-tuning requires modification of database configuration parameters
 - Some are changed in real time using SQL
 - Some affect database instance
 - Others affect entire RDBMS and all instances
- Initialization parameters reserve resources used by the database at run time
- After modifying parameters, may need to restart the database

Thank you for listening!
